Fiscal Year:	FY 2017	**Task Last Updated:**	FY 09/21/2017
PI Name:	Khokhlova, Vera Ph.D., D.Sc.		
Project Title:	Development and Validation of Therapeutic Ultrasound for Bloodless Surgery in Clinical Simulation in an Analog Environment		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Smart Medical Systems and Technology Team		
Joint Agency Name:		**TechPort:**	No
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Space Biology Element:	None		
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PI Email:	vera@apl.washington.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	206-221-6585
Organization Name:	University of Washington		
PI Address 1:	1013 NE 40th Street		
PI Address 2:	Applied Physics Laboratory		
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98105-6606	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-Crew Health (FLAGSHIP, NSBRI, OMNIBUS). Appendix A-Crew Health, Appendix B-NSBRI, Appendix C-Omnibus
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sapozhnikov, Oleg Ph.D., D.Sc. (University of Washington) Dunmire, Barbrina M.S. (University of Washington) Wang, Yak-Nam Ph.D. (University of Washington) Crum, Lawrence Ph.D. (University of Washington) Schade, George M.D. (University of Washington) Kreider, Wayne Ph.D. (University of Washington)		
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Task Description:	As astronauts venture farther from Earth, the risks of medical crises increase. Ultrasound is a well-known imaging modality that has been successfully used on the International Space Station (ISS) to help mitigate these risks. In addition, there is potential to leverage advances in Earth-based clinical applications of high intensity focused ultrasound (HIFU) to enable non-invasive surgical treatments during exploration missions. HIFU therapies have the potential to treat a wide range of indications that could occur in future exploration missions, including stopping internal bleeding with acoustic hemostasis, treating cancerous tumors that may arise quickly from exposure to radiation, palliation of pain, or even acting as an acoustic scalpel to treat serious internal injuries noninvasively, through the skin. Currently, the most advanced clinical HIFU systems use magnetic resonance imaging (MRI) to target and monitor a thermal effect of the treatment. New specialized HIFU approaches use sequences of high amplitude ultrasound pulses and low duty factors to generate shock waves and bubble activity at the targeted site. For space-based applications, these new protocols have particular appeal because bubbles are visible on ultrasound B-mode images. The flexible ultrasound system (FUS) designed to provide imaging capabilities on space exploration missions therefore can be used for targeting and monitoring such HIFU therapies. In addition, HIFU therapies that utilize shock waves have been shown to significantly accelerate treatments and create a diverse array of bioeffects ranging from pure thermal to pure mechanical (histotripsy). To realize the potential of using HIFU with FUS guidance in space, a suitable test-bed phantom is needed to provide a training apparatus for astronauts and test the delivery of therapy to targeted anatomical sites. The goal of this project was to develop such phantom with anatomical and acoustical features relevant to clinical HIFU such as skin, ribs, inhomogeneous body wall, and tissue targets to simulate thermal and mechanical HIFU treatments. For abdominal targets, the presence of ribs and inhomogeneous adipose tissue can affect shock formation through aberration, absorption, and diffraction. A HIFU transducer (1.2 MHz, 12.5 cm diameter and focal length) containing an imaging probe in its central opening was developed. A power source was built with the ability to deliver either short pulses at high powers (10 ms pulses at 1% duty cycle at up to 5 kW electrical power) for mechanical treatments or continuous output at moderate powers up to 700 W for thermal treatments. To represent fat and muscle in body wall, phantom layers were made from polyvinyl alcohol with irregular-shaped walls; the skin was represented by a silicone sheet; ribs were fabricated by 3D-printing based on a human anatomical model relevant to liver or kidney treatments. HIFU targets comprised optically transparent alginate or polyacrylamide gels that allows visualization of the lesion. These components were assembled in a water tank to provide coupling between layers and allow shifting of their relative positions. Free-field hydrophone measurements were performed for transducer characterization and shocked waveforms with peak positive/negative pressures of +100 / -20 MPa were measured; in situ fields behind ribs and body wall were then characterized both by hydrophone measurements and observations of lesion formation. When ribs were present, shocks formed at about half amplitude at the same power, and higher pressures were measured with ribs positioned closer to the transducer. The presence of a uniform body wall structure attenuated shock amplitudes by a smaller amount and was insensitive to axial position. Mechanically ablated lesions were visualized in target gels. These results showed that the presence of ribs and absorptive tissue-mimicking layers did not prevent shock formation at the focus and each layer simulated the impact on the beam focusing and shock formation similarly to the observations from in vivo and ex vivo experiments. Overall, with real-time lesion visualization, the developed low-maintenance HIFU phantom system compatible with FUS can serve as a training tool for researchers, engineers, and clinicians to facilitate faster acceptance of HIFU therapies for a number of applications to be used both in Earth-based and space-based conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is based on the using ultrasound to mechanically ablate tissue (histotripsy), noninvasively, through the skin. Histotripsy may have clinical advantages over traditional thermal HIFU including real-time ultrasound imaging of treatment, avoiding thermal injury of intervening tissues, rapid resorption of the fragmented tissue or its natural passage, and broader spectrum of potential bioeffects in tissue including ablation of tumors, lysis of venous thrombus, treating cardiac defects and brain pathology, enhanced immune response, or release of biomarkers. The same technology may be used to enhance thermal heating for faster tissue ablation or hemostasis application. Although the technology is at an early stage of development, it is already emerging for many clinical indications, and the Univ. of Washington team has a strong track record for supporting and commercializing their novel ultrasound technologies.
Task Progress:	The following tasks have been accomplished. • High intensity focused ultrasound (HIFU) test-bed phantom has been designed, fabricated, and tested. The phantom includes a water tank, high power source, a therapeutic HIFU transducer with a central opening to accommodate an imaging probe, an imaging probe connected to the Flexible Ultrasound System (FUS) for targeting and monitoring the treatment, phantoms of the body wall, ribs, and target phantoms for thermal and mechanical treatments. • Ultrasound therapeutic transducer and high power ultrasound source for generating both pulsed and continuous HIFU exposures have been designed in collaboration with Sonic Concepts and built by them. A transducer with driving electronics operates at 1.2 MHz with the ability to deliver either short pulses at high powers (10 ms pulses at 1% duty cycle at up to 5 kW electrical power) or continuous output at moderate powers up to 700 W (Sonic Concepts, Bothell, WA). • Free-field hydrophone measurements were performed for transducer characterization in water; it was shown that it is capable to generate shock waves higher than 100 MPa amplitude at the focus, which is sufficient to induce mechanical tissue ablation. • A body wall phantom consisting of gel layers with non-plane interfaces of irregular geometry to mimic inhomogeneous acoustic path in tissue has been prepared and tested. Representative rib phantoms have been designed to present clinically relevant geometries of sonicating liver and kidney and 3D printed from plastic material. These phantoms represent propagation paths in the body to the target site and introduce absorption, aberrations, and scattering of HIFU. In situ fields behind ribs and body wall were then characterized both by hydrophone measurements and observations of lesion formation. • Three types of the target phantoms have been prepared and tested. Two transparent gel phantoms of different type enable optical visualization of the mechanical exposures due to generation of bubbles combined with ultrasound imaging using FUS. It is shown that the appearance of the mechanical lesion and the size/shape of the lesion is similar to those observed in tissue. A thermochromic non-transparent gel phantom will be used for mimicking thermal treatments and exploring nonlinear heating regimes of HIFU.

	• Thermochromic inks and encapsulated inks have been tested to improve optical imaging of the ablated volumes in the transparent gels.
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